

FinTrust Digital Bank — Week 3 Data Analytics Project Summary

Programme: AnalystLab Africa Experience Lab Internship Programme

Track: Data Analytics

Project: FinTrust Financial Intelligence & Digital Banking Support Solution

Prepared by: Oluwayomi Precious Similoluwa

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1. Overview

Week 3 focused on extending the analysis completed in Week 2, testing previous findings and developing a clearer understanding of customer transaction value, monthly performance, channel activity and risk-review patterns. The goal was to move beyond descriptive analysis and use the available evidence to support more useful business interpretations.

2. Work completed or developed

SQL analysis

The Week 3 SQL analysis explored customer value within customer segments, monthly transaction value and growth, risk-review rates by transaction value band, customer-level average transaction size, customer contribution within segments, monthly channel performance, customer transaction frequency and transaction-value concentration.

Python analysis

The Week 3 notebook was designed to extend the earlier exploratory analysis with a customer-value concentration curve, monthly transaction-value trend, risk-review comparison by segment and value band, channel trajectories and a customer-frequency distribution.

The notebook currently has a CSV file-path error involving `cust_clean.csv`. Some outputs have been shared for review, but the complete notebook still needs to be rerun from the first cell before successful end-to-end execution can be confirmed.

Power BI dashboard

The existing Week 2 dashboard is being reviewed for improvements in layout, readability, filtering, KPI accuracy and interaction. The Week 3 version will be saved separately to preserve the original dashboard.

KPI reconciliation, slicer and interaction testing, final visual improvements and screenshots remain outstanding.

3. Key findings from the supplied SQL results

- The highest-value 25% of customers generated approximately ₦286.2 million, representing about 51% of total transaction value.
- High-value transactions had higher synthetic risk-review rates than normal transactions across all four customer segments.
- Monthly transaction value was approximately ₦188.49 million in January, ₦175.79 million in February and ₦196.20 million in March.
- ATM transaction value increased each month, while USSD transaction value decreased each month during the observed quarter.
- Most customers, 917 out of 1,500, fell into the six-to-ten transaction frequency group.
- A total of 605 customers had a customer-level average transaction amount above the platform-wide transaction-level average.

4. Validation and refinement

The Week 3 analysis provided a more detailed view of the earlier findings. Monthly transaction totals were compared across the quarter, and daily averages were considered to account for differences in the number of days in each month.

The channel analysis also distinguished between month-to-month growth and the overall change from January to March. Risk-review rates were compared across

transaction value bands and customer segments to determine whether the pattern was consistent.

The SQL results support these interpretations, but Python rerunning and Power BI reconciliation are still required before the validation process can be marked complete.

5. Business recommendations

The findings suggest exploring value-based customer engagement, monitoring transaction performance using daily averages, tracking channel performance over a longer period, examining the synthetic risk-review pattern carefully and testing engagement initiatives for different transaction-frequency groups.

These are recommendations for further investigation rather than claims about actual customer behaviour. The available data does not establish customer churn, profitability, fraud or long-term channel migration.

6. Development decisions

The work continues from the existing FinTrust project rather than introducing a new dataset or unrelated analysis. SQL results are being used to frame validation questions for Python and Power BI. Unusually large transactions are not automatically treated as errors, and the synthetic risk-review label is not interpreted as proof of fraud.

7. Assumptions, limitations and risks

The dataset is synthetic and covers Q1 2026 only. The high-value threshold used in the shared risk comparison is ₦117,922.86 and should be applied consistently. Monthly totals are affected by the number of days in each month, and low transaction frequency does not establish that a customer is inactive or dissatisfied.

The Python notebook has not yet been confirmed to run successfully from beginning to end. Power BI KPI reconciliation and interaction checks also remain outstanding.

8. Week 4 priorities

The next steps are to correct the Python file paths, rerun the notebook, reconcile results across SQL, Python and Power BI, complete dashboard improvements, capture evidence screenshots, and finalise the project documentation and repository.

9. Current project status

The Week 3 SQL analyses and draft interpretations are available, and the Python notebook has produced outputs shared for review. The project remains in progress while Python execution and Power BI validation are completed. Final submission materials should reflect the results of those checks rather than assume they have passed.

Findings & Validation

1. Purpose of the validation

The aim of Week 3 was not just to produce more analysis, but to check whether the findings from Week 2 still held when examined from different angles. I used additional SQL queries and planned Python visualisations to investigate customer transaction value, monthly performance, channel activity, customer frequency and risk-review patterns.

The findings below are based on the SQL results and Python outputs shared during the analysis. The Python notebook still needs to be rerun successfully from beginning to end, and the Power BI figures still need to be checked before final validation can be confirmed.

2. Findings and supporting evidence

Finding 1: A relatively small group of customers accounts for a large share of transaction value

The highest-value 25% of customers generated approximately ₦286.2 million out of the platform's total transaction value of about ₦560.5 million. This represents roughly 51% of the total value.

Evidence: SQL Q8, which divided customers into four groups based on their total transaction value, and the shared Python customer-value concentration output, which reported a top-quartile share of 51.0%.

Validation outcome: The two supplied outputs support the same general finding. The Python notebook should be rerun to confirm the result after its file paths are corrected.

What this means: Transaction value is not distributed evenly across customers. This may help the business explore different engagement approaches for different customer groups, although high transaction value alone does not establish profitability or loyalty.

Finding 2: High-value transactions have higher risk-review rates across customer segments

The SQL results showed risk-review rates of approximately 31–33% for high-value transactions, compared with approximately 17–19% for normal transactions. This pattern appeared across Everyday, Premium, SME and Student customers.

Evidence: SQL Q3 and the shared Python risk-review comparison by segment and value band.

Validation outcome: The pattern is consistent across the four segments in the supplied results. The notebook must be rerun, and the ₦117,922.86 threshold must be applied consistently when the results are reproduced.

What this means: Transaction amount is associated with the synthetic review label in this dataset. This does not prove that amount causes a review or that a flagged transaction represents fraud.

Finding 3: Monthly transaction value changed during the quarter

Total transaction value was approximately ₦188.49 million in January, ₦175.79 million in February and ₦196.20 million in March. February declined by 6.74% compared with January, while March increased by 11.61% compared with February.

Evidence: SQL Q2, which calculated monthly value, cumulative value and month-over-month percentage change.

Validation outcome: The supplied SQL figures support the reported changes. However, monthly totals should be considered alongside daily averages because the months contain different numbers of days.

Refinement: Using the supplied totals, average daily value was approximately ₦6.08 million in January, ₦6.28 million in February and ₦6.33 million in March. This suggests that February's lower total was partly related to its shorter calendar length, while the daily average increased modestly across the quarter.

Finding 4: Most customers fall within the medium-frequency group

The frequency analysis identified 293 low-frequency customers, 917 medium-frequency customers and 290 high-frequency customers. These represent approximately 19.5%, 61.1% and 19.3% of the 1,500 customers, respectively.

Evidence: SQL Q7, which grouped customers according to their number of transactions.

Validation outcome: The reported counts add up to 1,500 customers, consistent with the dataset's stated customer count.

What this means: Most customers made between six and ten transactions during the period. However, a low transaction count alone does not establish that a customer is inactive, dissatisfied or likely to leave.

Finding 5: Transaction value moved differently across channels

ATM transaction value increased from approximately ₦25.10 million in January to ₦28.77 million in February and ₦30.07 million in March. USSD transaction value

declined from approximately ₦14.67 million to ₦14.13 million and then ₦13.96 million.

Evidence: SQL Q6, which compared monthly transaction value across channels.

Validation outcome: The supplied figures show ATM value increasing and USSD value decreasing across the three months.

Refinement: ATM value increased by approximately 19.8% from January to March. The previously reported 14.6% increase refers specifically to January to February.

What this means: The channels followed different patterns during the observed period. Three months of synthetic data are not enough to establish a long-term trend or prove that customers are switching from USSD to other channels.

Finding 6: The overall average transaction amount may not represent every customer

The SQL result showed that 605 out of 1,500 customers, or approximately 40.3%, had an average transaction amount above the platform-wide transaction average of about ₦46,706.

Evidence: SQL Q4, which compared each customer's average transaction amount with the overall transaction-level average.

Validation outcome: The reported customer count corresponds to approximately 40.3% of the customer base.

What this means: The overall mean may not be representative of every customer's typical transaction. The median should also be calculated and compared with the mean before deciding which measure best represents a typical transaction.

3. Improvement-and-retest cycle

Initial output: Week 2 described transaction value as relatively steady and examined channel performance over the quarter.

Test: Week 3 SQL analysis compared monthly transaction value, calculated month-over-month changes and examined individual channel trends.

Finding: February's total transaction value declined by 6.74%, March's increased by 11.61%, ATM value rose each month, and USSD value declined each month.

Refinement: I refined the interpretation by considering daily averages and correcting the distinction between ATM's month-to-month growth and its overall January-to-March growth. I also avoided treating a three-month pattern as a long-term forecast.

Re-test required: Rerun the Python notebook from the first cell, confirm that the results match SQL, and reconcile the relevant figures with the Power BI dashboard.

Current result: The supplied SQL evidence supports a more detailed interpretation than the original broad description of a steady trend. The full validation cycle remains open until the Python and Power BI checks are completed.

4. Outstanding validation tasks

- Correct the Python CSV file paths and run all notebook cells successfully.
- Confirm that the CSV files and SQL database contain the same cleaned data.
- Reconcile the six Power BI KPI cards with the dataset and SQL results.
- Test date filtering, slicers and visual interactions.
- Save genuine screenshots of the successful Python outputs and improved dashboard.

5. Data limitations

The FinTrust dataset is synthetic and covers January to March 2026. The results should therefore be treated as educational findings rather than evidence of actual customer behaviour. In particular, the risk-review flag is a synthetic label and must not be described as proof of fraud. The analysis also cannot establish long-term trends, causation, customer churn or profitability without additional evidence.

Business Recommendations

1. Explore customer engagement based on transaction value

Finding: The highest-value 25% of customers generated approximately 51% of total transaction value.

Evidence: SQL Q8 and the shared Python customer-value concentration output.

Business implication: Transaction value is concentrated among a portion of the customer base, so understanding the needs and activity of these customers may help the business make better engagement decisions.

Recommendation: Explore a value-based customer engagement approach that provides appropriate support and relevant product information to different customer groups. Monitor the results before expanding the approach.

Limitation: High transaction value does not automatically mean a customer is profitable, loyal or likely to leave.

2. Examine the factors associated with risk reviews

Finding: High-value transactions had higher synthetic risk-review rates than normal transactions across all four customer segments.

Evidence: SQL Q3 and the shared Python comparison.

Business implication: The synthetic review label appears more frequently in the high-value transaction group.

Recommendation: Document how the review flag and high-value threshold are defined. Examine the pattern alongside other relevant transaction characteristics before considering how such analysis might inform a real review process.

Limitation: These results show an association, not causation. The synthetic flag does not establish fraud or demonstrate how a real bank should make risk decisions.

3. Compare monthly performance using daily averages

Finding: Total transaction value declined in February and increased in March, but monthly totals are affected by the number of calendar days.

Evidence: SQL Q2 and the monthly totals supplied for January, February and March.

Business implication: Looking only at total monthly value can make performance changes harder to interpret.

Recommendation: Present monthly transaction value alongside average daily transaction value. Continue monitoring over a longer period before making decisions based on apparent growth or decline.

Limitation: The available three-month period is too short to establish seasonality or support a reliable forecast.

4. Monitor channel performance before changing priorities

Finding: ATM transaction value increased each month, while USSD transaction value declined during the same period.

Evidence: SQL Q6.

Business implication: Channel activity may be changing, but the reasons behind those changes are not established by the figures alone.

Recommendation: Track transaction count, transaction value and success rates by channel over subsequent periods. If real customer feedback becomes available, use it to investigate possible reasons for the differences.

Limitation: The analysis does not prove that customers are migrating from USSD to mobile or web channels, so it would be premature to recommend withdrawing support for USSD.

5. Tailor engagement experiments to transaction frequency

Finding: Approximately 61.1% of customers made between six and ten transactions, while approximately 19.5% made between one and five.

Evidence: SQL Q7.

Business implication: Customers displayed different levels of transaction activity during the quarter.

Recommendation: Consider testing suitable onboarding, service education or engagement initiatives for different frequency groups. Compare the outcomes against a clear baseline before deciding whether an initiative is effective.

Limitation: Low transaction frequency does not necessarily mean inactivity or dissatisfaction. Customer needs and account circumstances should also be considered.

6. Improve how the typical transaction amount is reported

Finding: Only 40.3% of customers had a customer-level average transaction amount above the overall transaction-level average of approximately ₦46,706.

Evidence: SQL Q4.

Business implication: A single average may not fully represent the transaction experience across the customer base.

Recommendation: Calculate and validate the median transaction amount, then consider displaying the mean and median together on the dashboard with clear labels.

Limitation: The median value should not be published until it has been calculated from the transaction-level data and checked against the other analysis outputs.

Suggested priorities for Week 4

1. Complete Python and Power BI validation.
2. Reconcile key figures across SQL, Python and Power BI.
3. Improve the dashboard's customer-value and monthly-performance views where they add value without overcrowding the report.
4. Document the synthetic risk-review label and the high-value threshold clearly.
5. Carry the validated findings, recommendations and limitations into the final project report.